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Calibration of SNO+ using the LaserBall

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The SNO+ Experiment:

Physics experiment with the following goals of measuring:

$0\nu\beta\beta$

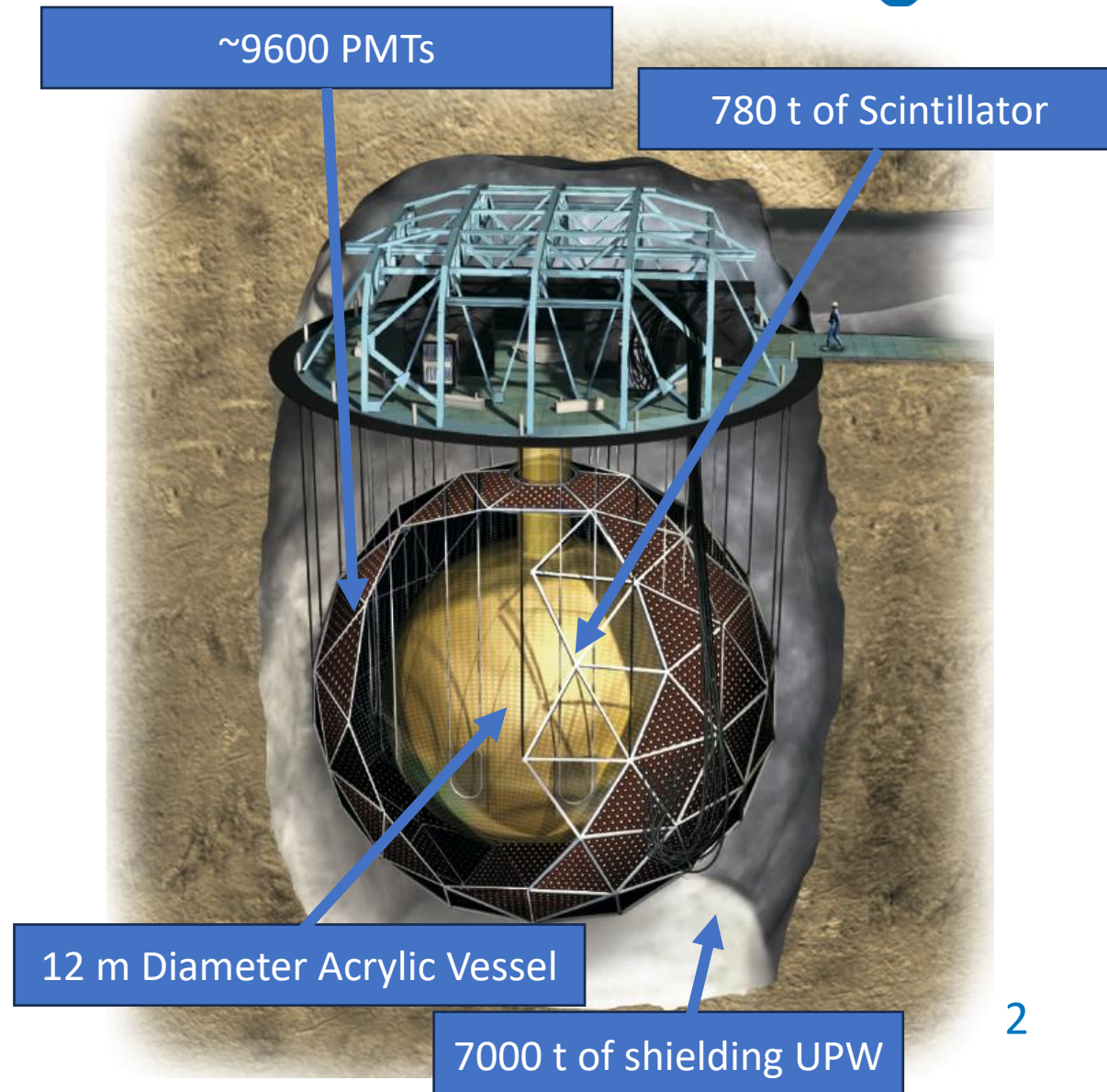
Solar Neutrinos

Geoneutrinos

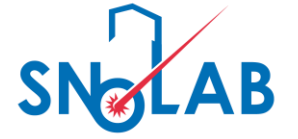
Reactor Antineutrinos

Supernova Neutrinos

Nucleon Decay



Calibrating the SNO+ Detector



Goal:

Validate detector response at key energies

Validate position reconstruction

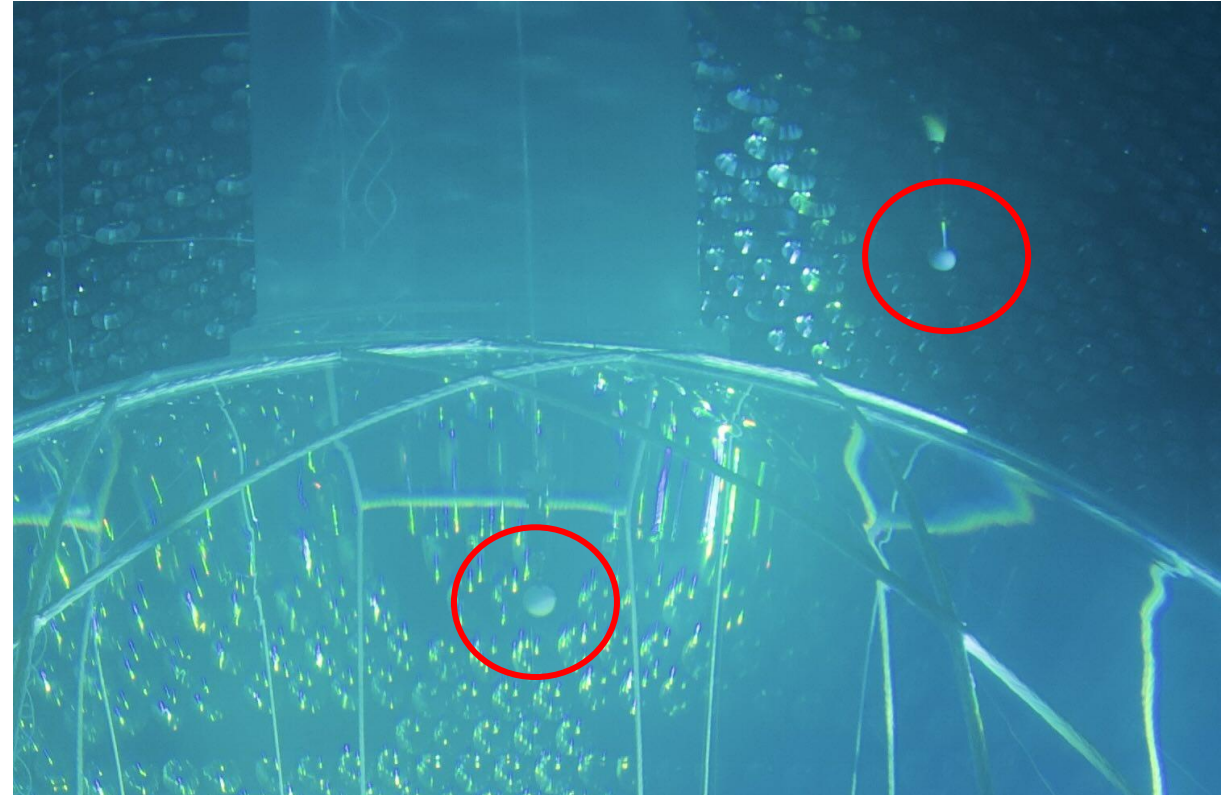
Determine PMT timing and response

Why:

Needed to ensure detector data is accurate

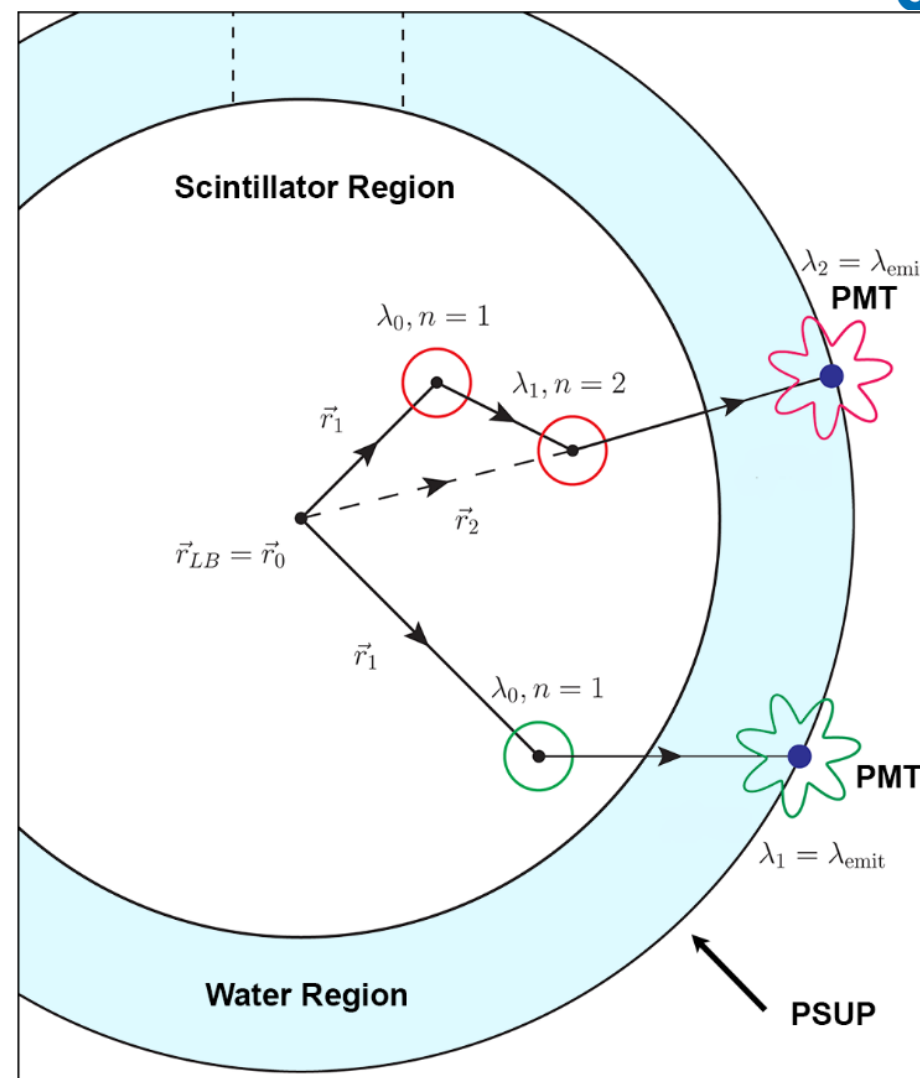
Method:

Deploying radiation sources & optical sources



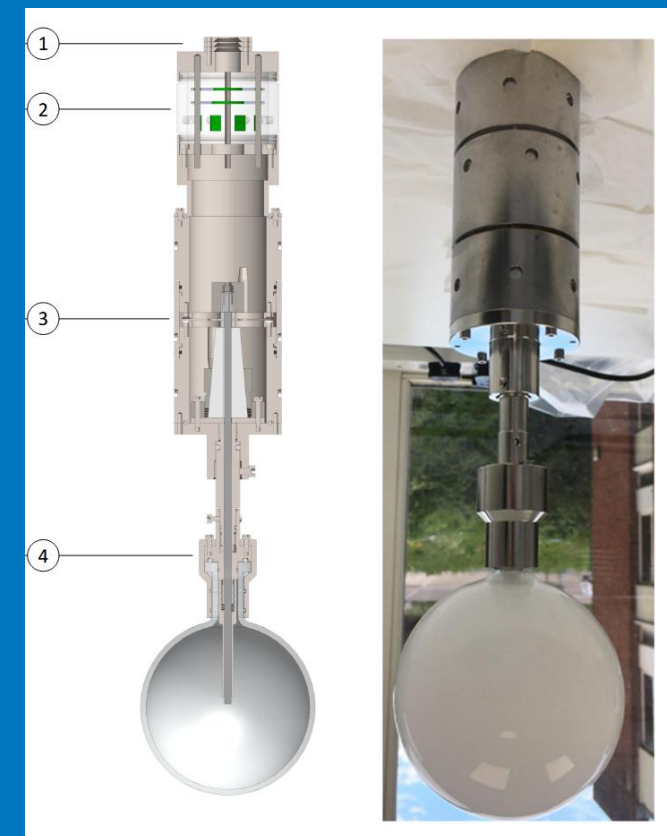
Photomultiplier Tubes

- Arranged symmetrically in a geodesic sphere
- PSUP is 19 meters across
- Most sensitive from 300 nm to 650 nm wavelength



What is the LASERBALL?

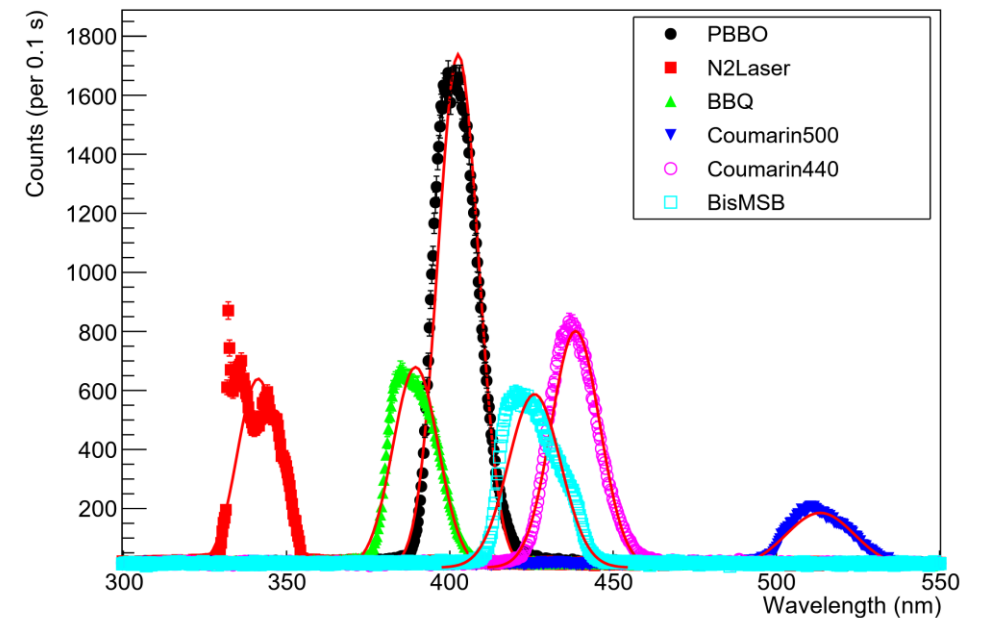
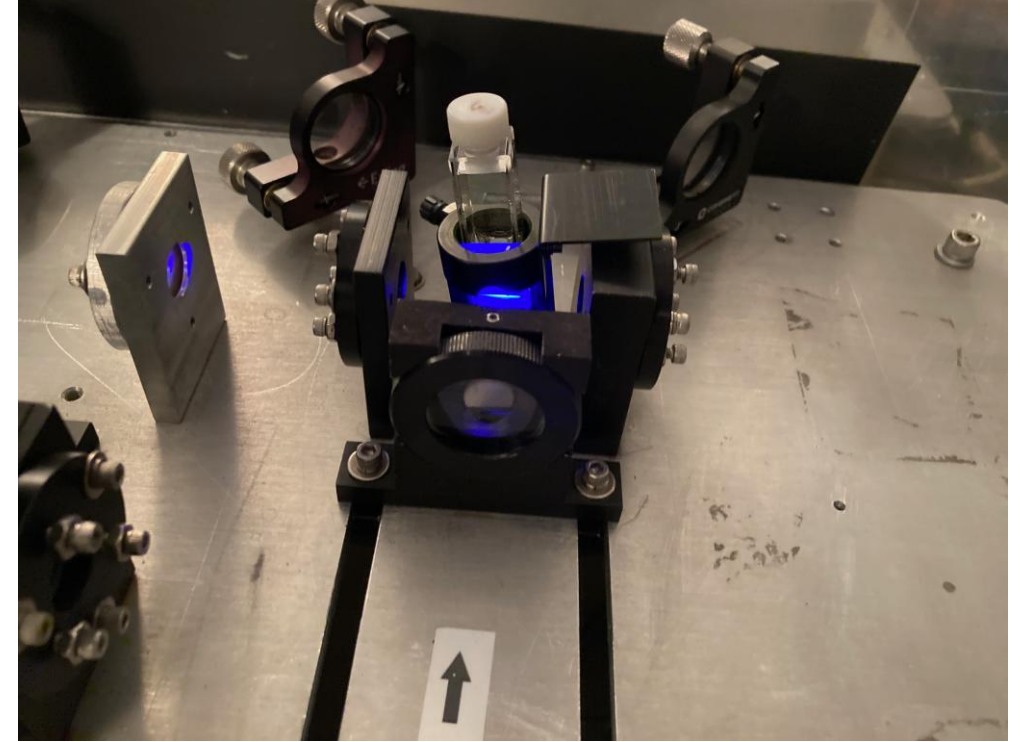
- Multi-Wavelength, isotropic laser source
- Used to measure relative PMT efficiencies, timing & accuracy
- Connected to external laser source, emitting 6 different wavelengths



1. O-Ring gasket stack for umbilical seal
2. Umbilical Flasher Object
3. Source Connector
4. Laserball Calibration source

Dye Laser System

- Uses a Nitrogen (N₂) laser as the primary source
- Laser fired into dye cells generates secondary light
- Probes response across a wide spectrum of light

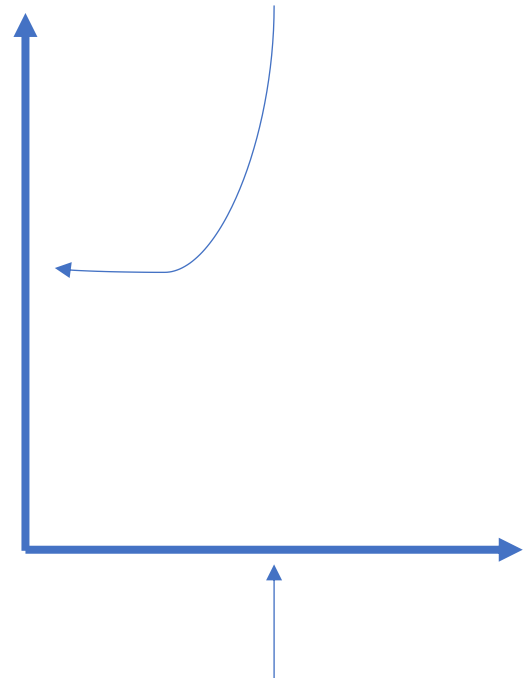




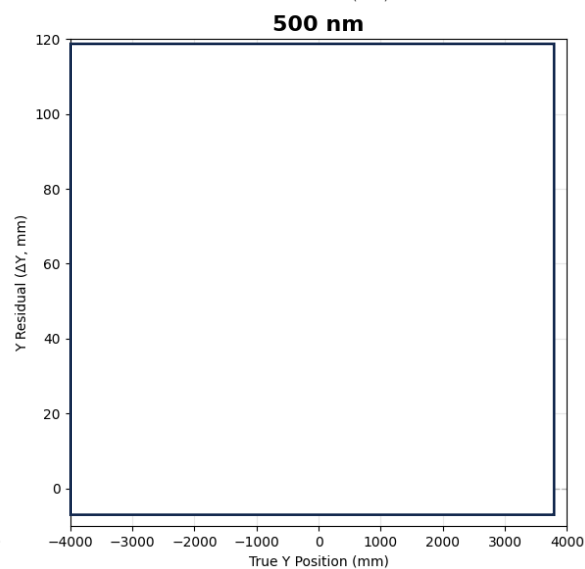
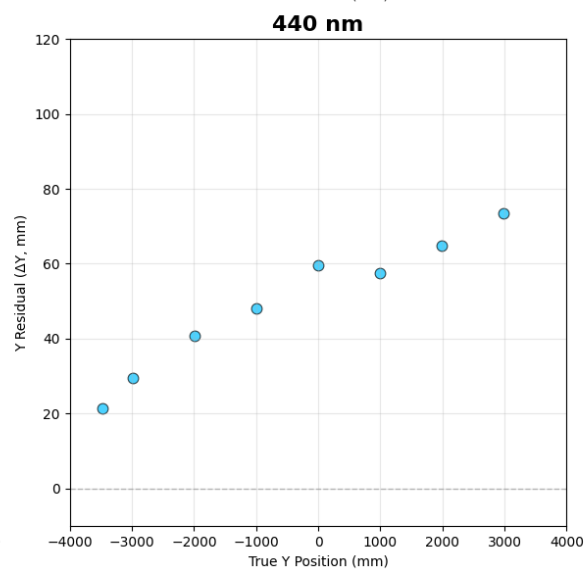
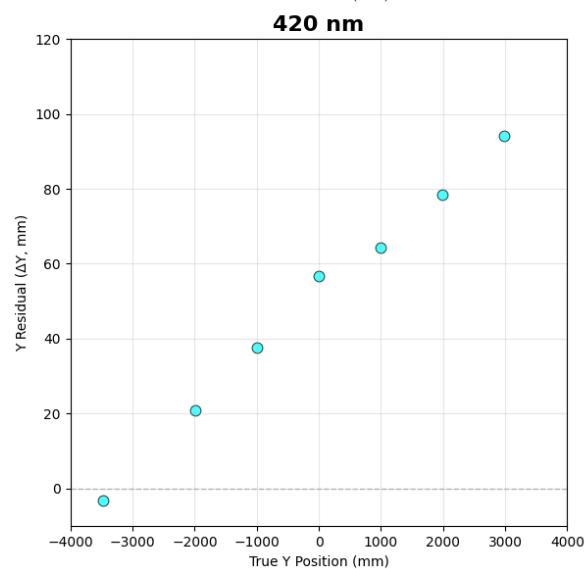
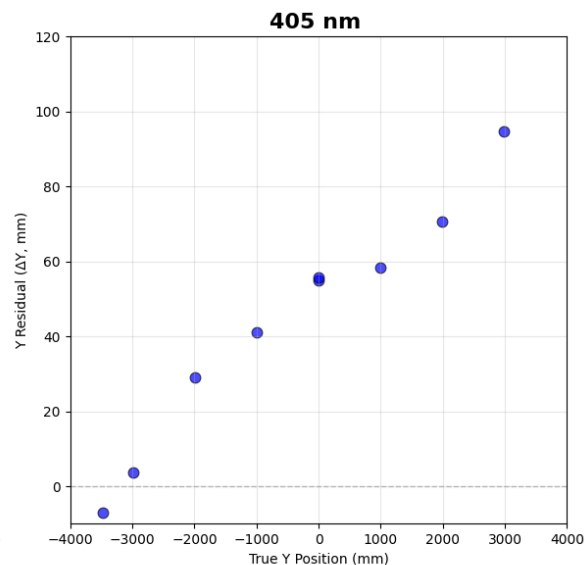
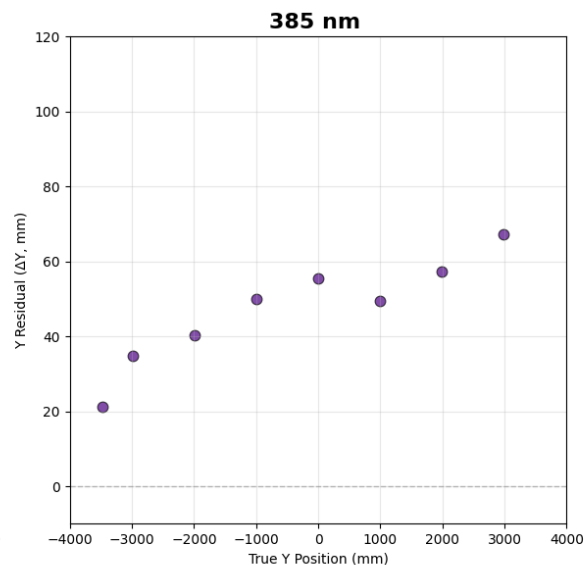
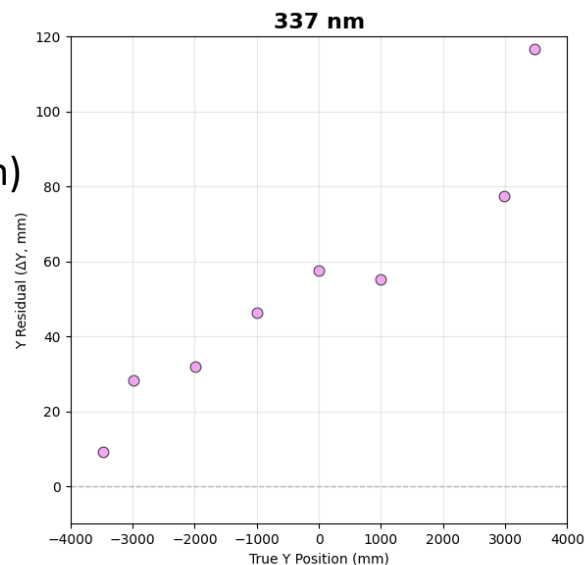
Reconstructing Laserball Position



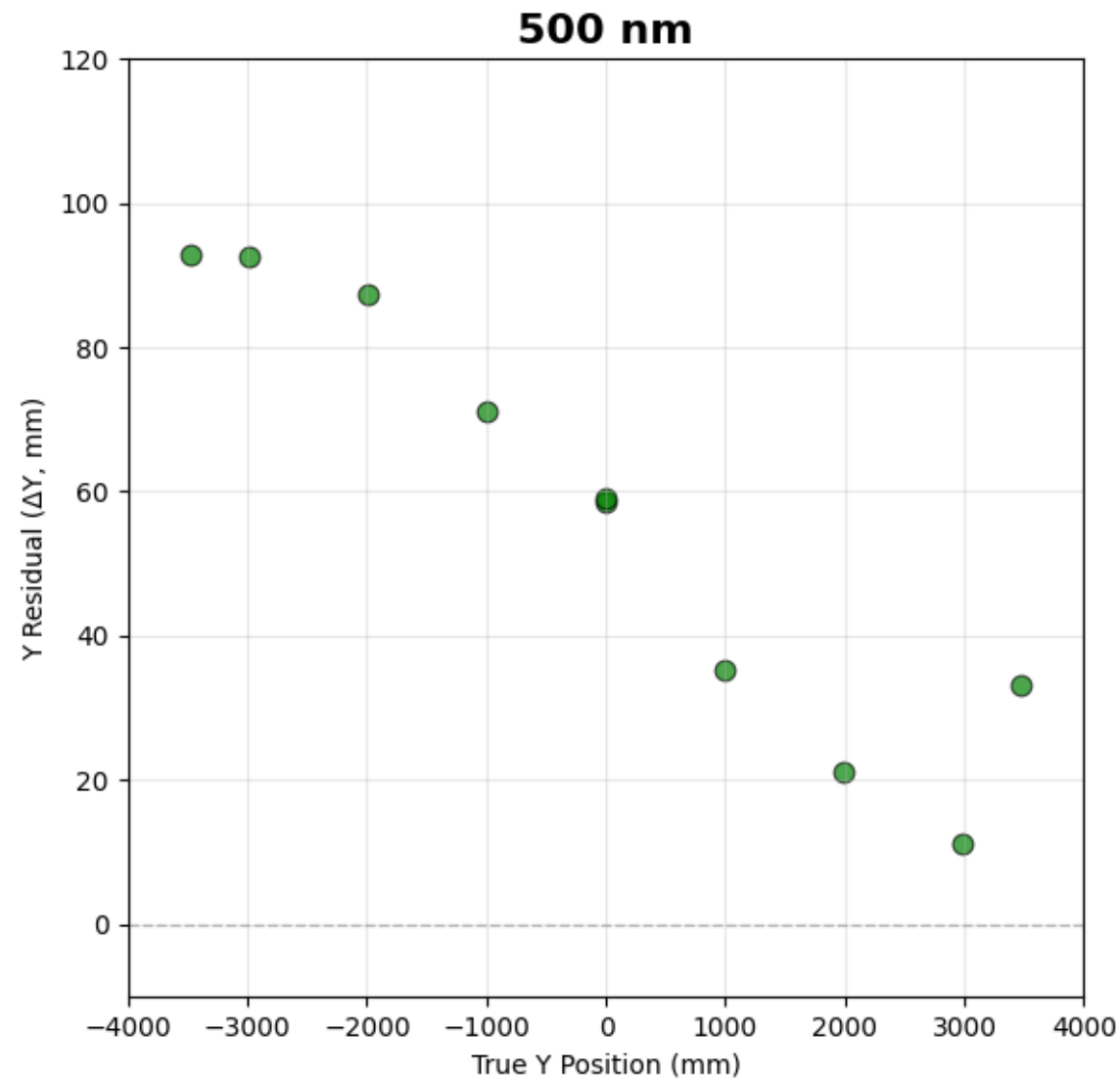
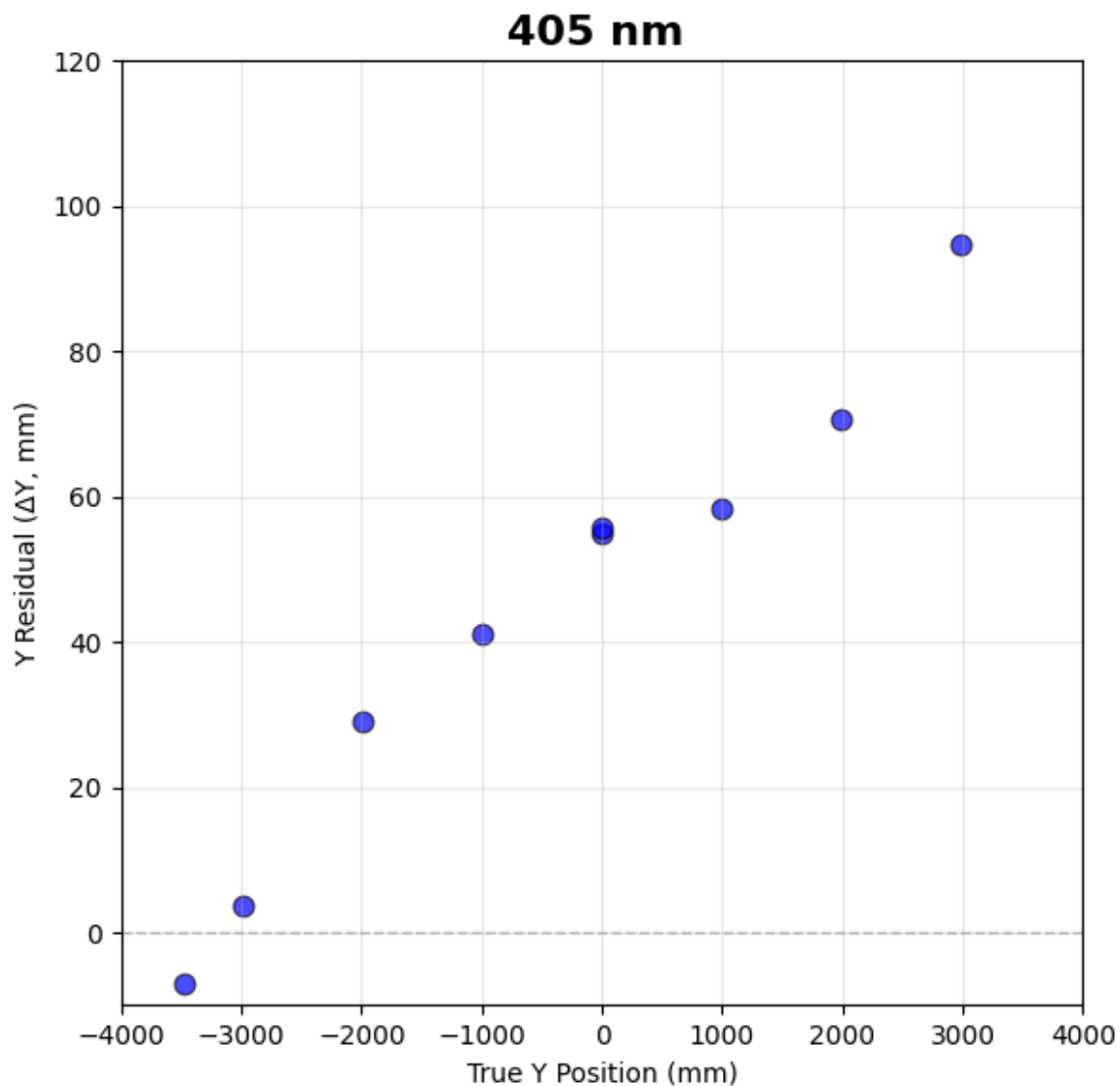
Y Residual
(Reconstr. Y Position – True Y Position)



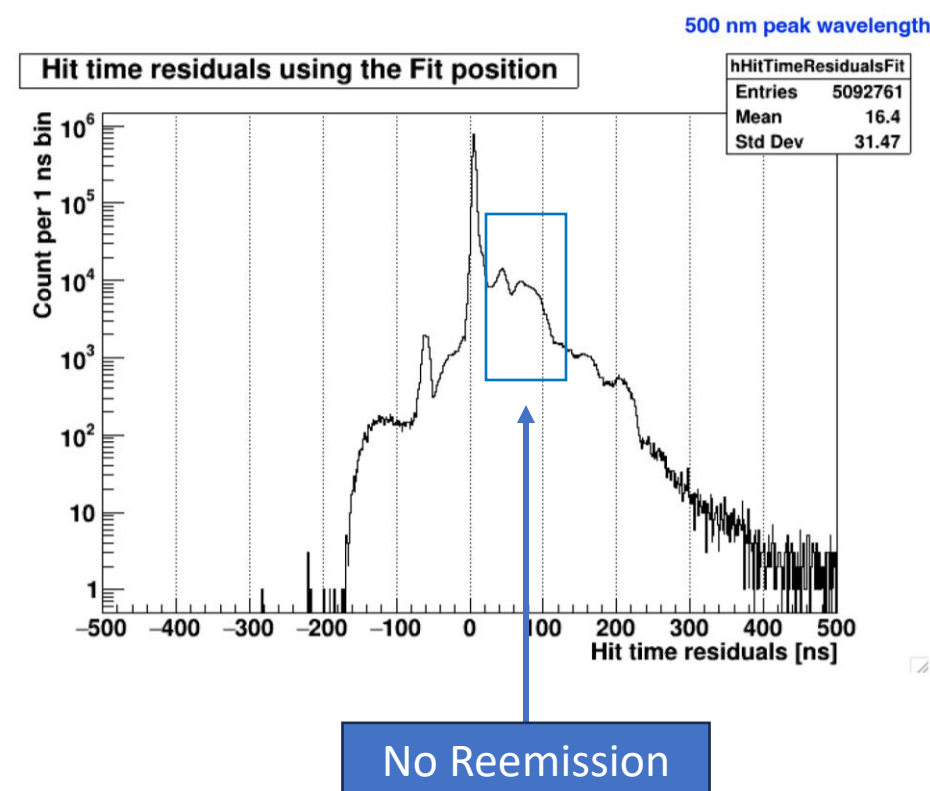
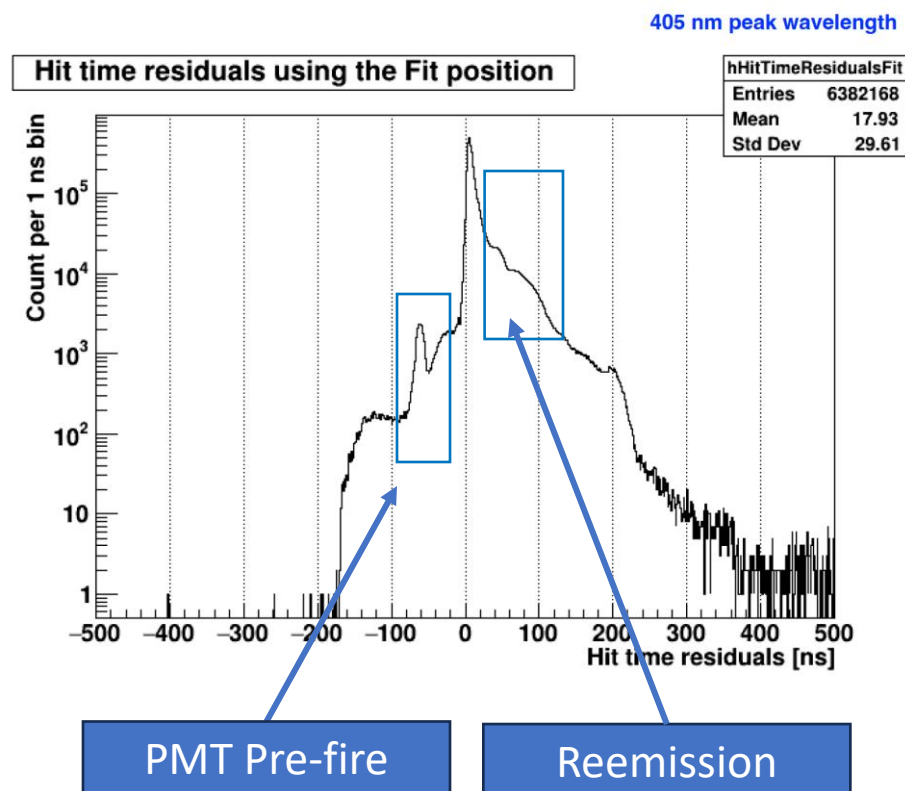
True Y Position



Reconstructing Laserball Position

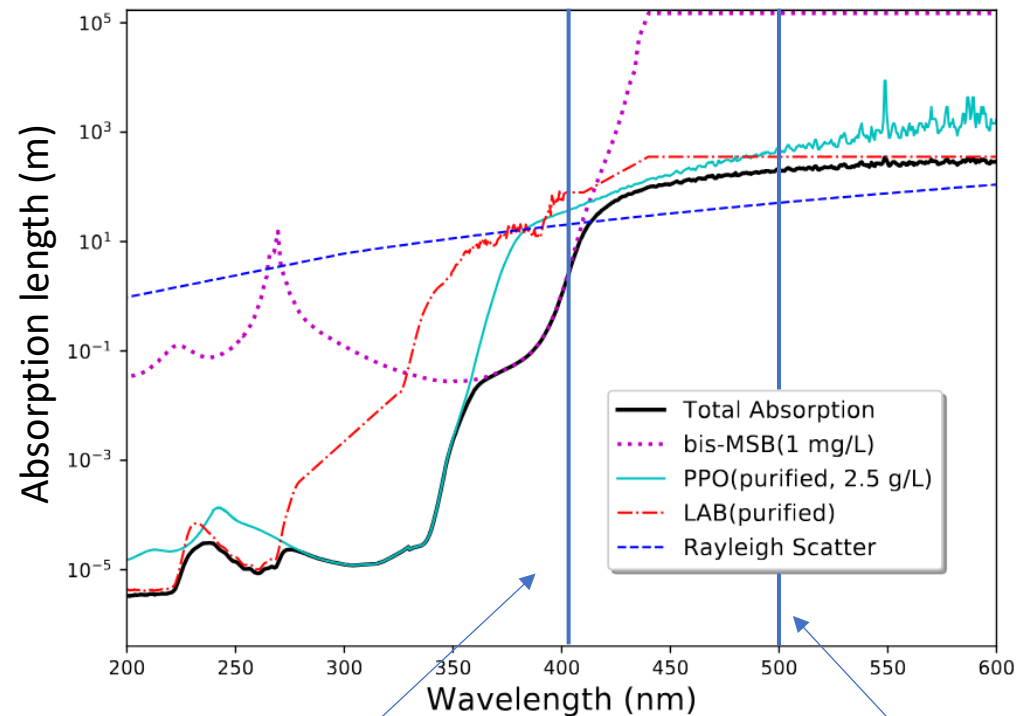


Timing Properties @ $\lambda=405$ nm and $\lambda= 500$ nm



Scintillator behavior as a function of λ

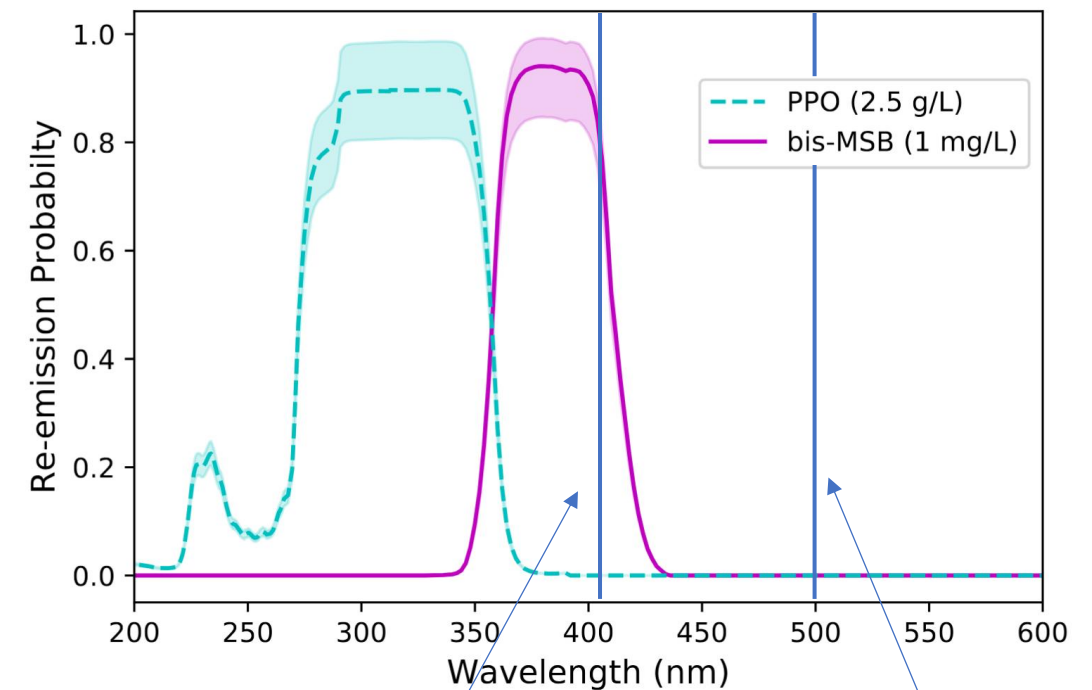
Absorption Plot



405 nm

500 nm

Reemission Plot



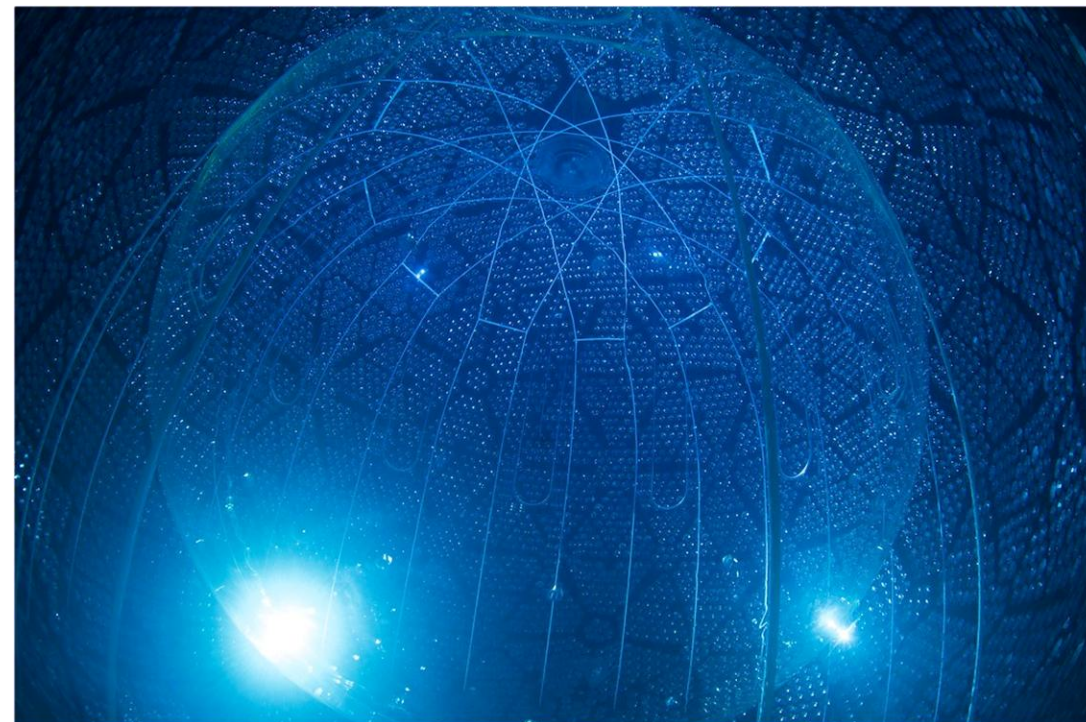
405 nm

500 nm

Conclusion

SNO+ in process of calibrating scintillator phase

- Observed positional bias due to change in wavelength
- Importance in understanding optics
- Deployment is finished, now we analyse the data



Thank you!
Questions?